

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Mid-Term Examination, 2025-26 (NEP)

MATHEMATICS**Paper Code: MATMJ-32 : Analysis (Mid Term)**

Time: 1 Hour

Maximum Marks: 20

Instructions: Attempt any 5 questions. All questions carry equal marks (4 marks each).

Question 1. What are Peano's Axioms? State and prove the Law of Trichotomy for $\mathbb{N}$, OR prove that $m \cdot (n + p) = m \cdot n + m \cdot p$ for all $m, n, p \in \mathbb{N}$. [4]

Question 2. What are the common algebraic and order properties shared between the set of real numbers $\mathbb{R}$ and the set of rational numbers $\mathbb{Q}$? Mention the completeness property (Least Upper Bound Axiom) that distinguishes $\mathbb{R}$ from $\mathbb{Q}$. Illustrate this difference with a concrete example. [4]

Question 3. What do you mean by a convergent sequence? Prove that a bounded sequence (x_n) is convergent if and only if $\limsup x_n = \liminf x_n$. [4]

Question 4. Differentiate between an absolutely convergent series and a conditionally convergent series. Provide standard examples for both types. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{\sin n\theta}{n^2}$. [4]

Question 5. If $f: [0, a] \rightarrow \mathbb{R}$ is defined by $f(x) = x^2$ with $a > 0$, prove that $f \in \mathcal{R}[0, a]$ (Riemann integrable) and evaluate $\int_0^a f(x) dx = \frac{a^3}{3}$. [4]

Question 6. Prove Cauchy's Criterion for Integrability: A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if and only if for any $\epsilon > 0$, there exists a partition P of $[a, b]$ such that $U(P, f) - L(P, f) < \epsilon$. [4]

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